

Boeing–Spirit: what was the collar worth?

All-stock merger · signing date 30 June 2024

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VALUE ON SIGNING DAY	COLLAR VS FIXED RATIO	CHANCE BELOW \$37.25	DELIVERED, 8 DEC 2025
\$36.83	−\$0.48	25% vs 48%	\$40.33

01 Summary

On 1 July 2024 Boeing announced the acquisition of Spirit AeroSystems, its largest supplier, for **\$37.25 per share paid entirely in Boeing stock**, a 30% premium to Spirit's \$28.60 close before reports of talks. The number of Boeing shares was not fixed: it adjusted so that Spirit holders received \$37.25, but only while Boeing's 15-day average price stayed between \$149.00 and \$206.94. Outside that band the ratio froze and Spirit holders carried Boeing's price risk again. The deal closed seventeen months later, on 8 December 2025.

This memo takes the seat of an advisor to the Spirit board on 30 June 2024. It prices the collar as a package of options on Boeing, using Boeing's implied volatility surface rebuilt from real option quotes, compares it with the two plain alternatives, and follows the offer every trading day to closing. On signing day the offer was worth **\$36.83: \$0.48 less than a fixed exchange ratio**, in return for cutting the chance of receiving less than \$37.25 from 48% to 25%. The band was tilted towards Boeing, and it was tested: Boeing closed below the floor on 16 days and touched \$136.59 in April 2025, before recovering. Spirit holders finally received **0.1955 Boeing shares, worth \$40.33**.

CONCLUSION
Accept the structure: the collar was fairly priced insurance on a volatile acquirer. But negotiate the band. A cap near \$201 instead of \$206.94 would have made the collar worth as much as a fixed ratio, and a price-linked walk-away below \$149 would have covered the risk the floor left open.

02 The terms

TERM	WHAT IT SAYS	COMMENT
Headline	\$37.25 per Spirit share in Boeing common stock	Merger agreement, 30 June 2024
Inside the band	Exchange ratio = \$37.25 / Boeing VWAP	Value fixed at \$37.25
Floor	VWAP at or below \$149.00: ratio fixed at 0.2500	18% below Boeing's \$182.01 close on 28 June
Cap	VWAP at or above \$206.94: ratio fixed at 0.1800	14% above the same close
VWAP window	15 trading days ending two days before closing	Shares delivered at closing
Conditions	Sale of the Airbus work packages; antitrust; Spirit vote	Closing expected mid-2025
Equity value	About \$4.7bn (\$8.3bn with net debt)	Roughly 126 million Spirit shares

Source: Spirit AeroSystems definitive proxy statement (DEFM14A, 20 December 2024) and the announcement of 1 July 2024.

03 What the offer was worth on signing day

Inside the band a Spirit share receives exactly \$37.25; below the floor it holds 0.25 Boeing shares, above the cap 0.18. That payoff equals \$37.25 in cash at closing, **minus 0.25 puts** on Boeing struck at \$149.00, **plus 0.18 calls** struck at \$206.94. Each option is priced at its own implied volatility from Boeing's surface on 28 June 2024, with closing expected at the end of June 2025.

COMPONENT	INPUTS	PER SPIRIT SHARE
\$37.25 paid at closing, discounted	4.96% for 1.01 years	\$35.44
0.25 puts on Boeing at \$149.00, sold by Spirit holders	vol 33.6%, price \$7.07	-\$1.77
0.18 calls on Boeing at \$206.94, kept by Spirit holders	vol 32.0%, price \$17.60	+\$3.17
Averaging and delivery lag (Monte Carlo)	15-day VWAP, shares two days later	-\$0.001
Value of the offer		\$36.83

STRUCTURE, SAME \$37.25 HEADLINE	VALUE TODAY	VS FIXED RATIO	CHANCE BELOW \$37.25	WORST 5%	MEDIAN	BEST 5%
Collar (actual terms)	\$36.83	-\$0.48	25%	\$25.58	\$37.25	\$55.03
Fixed exchange ratio, 0.2047	\$37.31	–	48%	\$20.94	\$37.72	\$62.57
Fixed value, no collar	\$35.44	-\$1.88	0%	\$37.25	\$37.25	\$37.25

Figures are rounded to the cent; totals and differences are computed before rounding: $\$35.4369 - \$1.7685 + \$3.1674 - \$0.0014 = \$36.8344$.

A fixed ratio of 0.2047 shares was worth \$37.31: Boeing pays no dividend, so shares received later are worth their price today. A fixed \$37.25 paid a year later was worth only \$35.44. The collar sat in between. It cost \$0.48 a share against the fixed ratio, about \$60 million in total, and in exchange lifted the worst 5% of outcomes from \$20.94 to \$25.58. Outcome ranges come from the risk-neutral distribution implied by the surface.

04 Could the board have negotiated a better band?

The band was not centred on Boeing's price. The floor gave Spirit holders protection only for the first 18% of a fall, while the cap handed Boeing everything above a 14% rise. A lower floor or a lower cap both add value for Spirit holders. The table shows alternatives with the same \$37.25 headline, priced on the same surface.

BAND	VALUE ON SIGNING DAY	VS ACTUAL
Actual: floor \$149.00, cap \$206.94	\$36.83	–
Floor lowered to \$140.00	\$37.24	+\$0.40
Cap lowered to \$200.00	\$37.40	+\$0.57
Symmetric ±15% around \$182.01	\$36.37	-\$0.47
Cap at \$201.02 (floor \$149)	\$37.31	+\$0.48
Floor at \$138.16 (cap \$206.94)	\$37.31	+\$0.48

A symmetric ±15% band, often the first draft in these negotiations, would have been worth less than the actual terms. The two levers that mattered were the cap, which sets how much upside Spirit gives away, and the depth of the floor. The implied distribution already gave Boeing a 25% chance of ending below \$149 by mid-2025: that is where a walk-away or re-pricing right would have been worth asking for.

05 What happened

DATE	EVENT
13 Sep 2024	Machinists strike begins (33,000 workers)
28 Oct 2024	Boeing prices about \$21bn of new equity at \$143
14 Nov 2024	Boeing closes at \$138.14, below the floor
31 Jan 2025	Spirit shareholders approve the merger
4 Apr 2025	Tariff crash: Boeing low of \$136.59
15 Oct 2025	EU approval, with the Airbus and Malaysia divestitures
3 Dec 2025	FTC consent order
8 Dec 2025	Closing: 0.1955 Boeing shares per Spirit share

The protection was tested in autumn 2024 and again in spring 2025: Boeing closed below \$149 on 16 of 362 trading days, and above the \$206.94 cap on 98. Realised volatility over the period was 36.9%, against about 32% implied at signing. Spirit traded on average \$1.33 below the model value of the offer; if a break meant a return to \$28.60, that spread implied a median 85% chance of completion.

OUTCOME AT CLOSING, 8 DECEMBER 2025	BOEING SHARES	VALUE	VS \$37.25
Collar (actual)	0.1955	\$40.33	+\$3.08
Fixed exchange ratio	0.2047	\$42.22	+\$4.97
Fixed value, no collar (VWAP inside the band: same ratio)	0.1955	\$40.33	+\$3.08
Collar, had the deal closed on 4 April 2025	0.2155	\$29.44	-\$7.81

Boeing closed at \$206.27 on the closing day. The ratio is set on the 15-day VWAP (about \$190.54; rebuilt here at \$190.74 from daily data) and the shares are delivered two days later, which is why a crash inside that gap, as on 4 April 2025, is not covered by the floor.

06 Recommendation

- **Accept the collar.** At \$0.48 a share it was fair insurance on an acquirer coming out of the door-plug crisis, and it halved the chance of receiving less than the headline.
- **Negotiate the band.** Ask for a cap near \$201, or a floor near \$138; either makes the collar worth as much as a fixed ratio.
- **Cover the tail the floor leaves open.** A walk-away or re-pricing right if Boeing's VWAP falls well below \$149, and a shorter gap between the VWAP window and delivery.

07 Limitations

- Listed Boeing options in the dataset reach only about seven weeks, so the one-year volatility is extrapolated flat. With any flat vol from 25% to 45% the collar stays \$0.23 to \$0.45 below a fixed ratio.
- Daily marks use the actual closing date; at signing closing was expected in mid-2025, which moves the value by about \$0.2 a share. The break price behind the implied probability is an assumption.
- The model does not price Spirit's own risks during the wait: the Airbus carve-out, regulatory remedies and Spirit's funding needs.

Sources. Boeing option chains and Boeing and Spirit prices: DoltHub (post-no-preference/options, /stocks). Treasury yields: FRED. Terms: Spirit DEFM14A (20 Dec 2024) and Form 8-K (8 Dec 2025). Events: Boeing, Spirit, European Commission and FTC releases. Method: Gatheral & Jacquier (2014), Breeden & Litzenberger (1978). Educational project, not investment advice.